MNIST Digit Classification using CNN and ANN

CS23B2043 Shishir Kumar Reddy Ambala
CS23B1004 Anumalasetty Sohan Kumar

1 Dataset

The MNIST dataset consists of 60,000 training and 10,000 test grayscale images of handwritten digits (28×28 pixels).

- CNN input shape: (60000, 1, 28, 28)
- ANN input shape: (60000, 784)
- Labels: One-hot encoded (60000, 10)

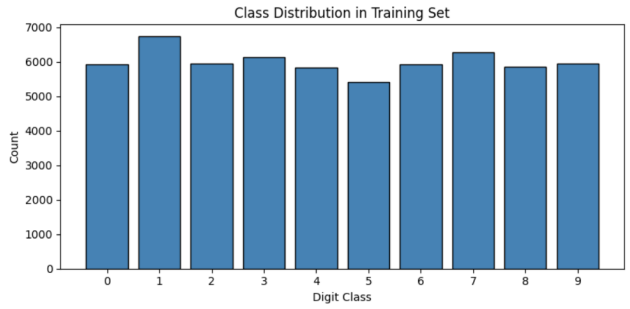

Figure 1: Class distribution

2 Model Architectures

2.1 CNN Architecture

- Conv(8, 3×3) + ReLU
- MaxPool(2×2)
- Conv(16, 3×3) + ReLU
- MaxPool(2×2)
- Flatten

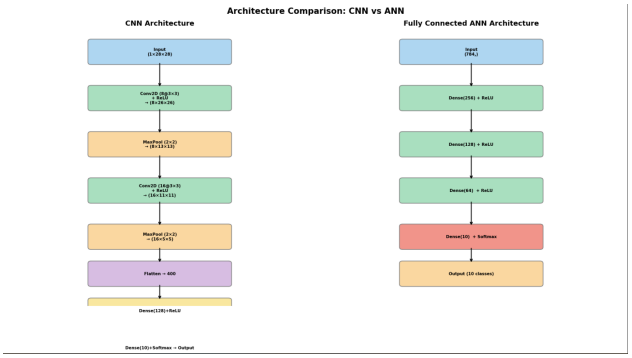

Figure 2: Architecture

- Dense(128) + ReLU
- Dense(10) + Softmax

Total parameters: ∼53,866

2.2 Fully Connected ANN

- Dense(256) + ReLU
- Dense(128) + ReLU
- Dense(64) + ReLU
- Dense(10) + Softmax

Total parameters: 242,762

3 Training Results

3.1 CNN Training Summary

- Epochs: 5
- Final Training Accuracy: 96.87%
- Test Accuracy: 96.91%
- Average Time per Epoch: ∼138s

Loss decreased from 0.5483 to 0.1074, indicating stable convergence.

3.2 ANN Training Summary

- Epochs: 10
- Final Training Accuracy: 94.99%
- Test Accuracy: 95.01%
- Average Time per Epoch: ∼4s

Loss decreased from 1.2799 to 0.1780, showing slower convergence relative to CNN.

4 Quantitative Comparison

Indicator	**CNN**	**ANN**
Number of Parameters	∼54K	∼243K
Input Representation	2D (28×28)	Flattened (784)
Training Accuracy	96.87%	94.99%
Test Accuracy	96.91%	95.01%
Epochs Trained	5	10

Table 1: CNN vs ANN Performance Comparison

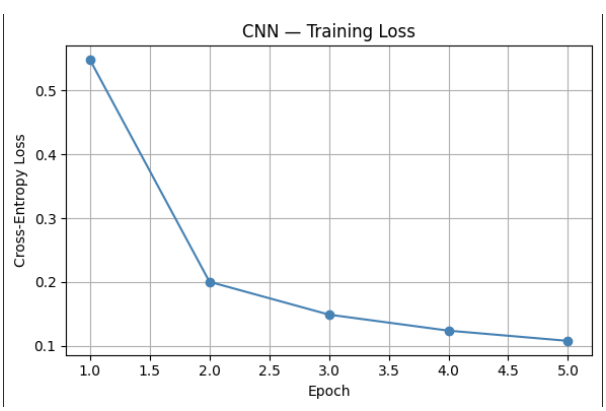

Figure 3: CNN Training Loss

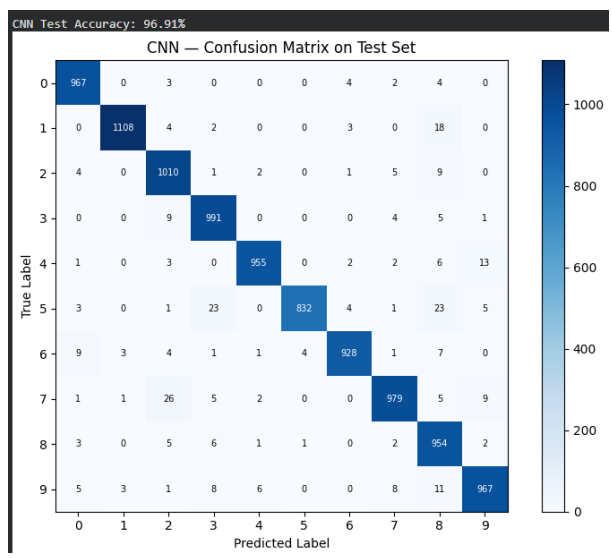

Figure 4: CNN Confusion Matrix on Test Dataset

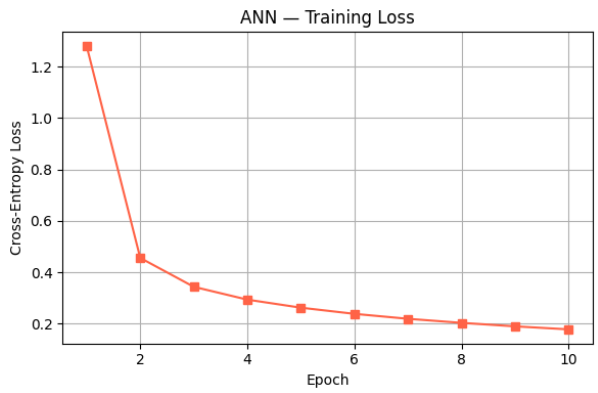

Figure 5: ANN Training Loss

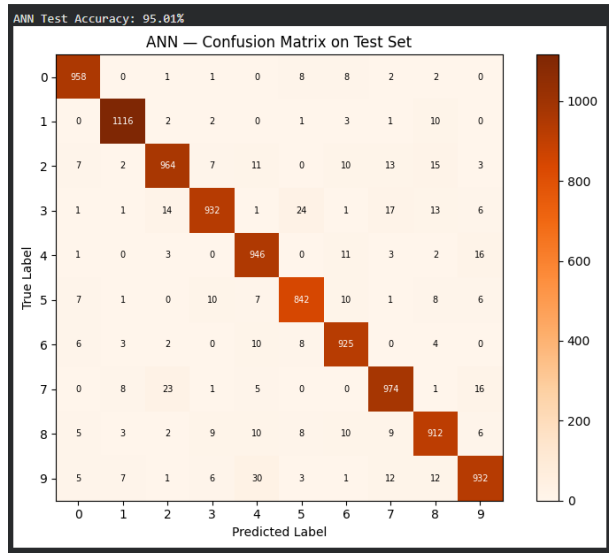

Figure 6: ANN Confusion Matrix on Test Dataset

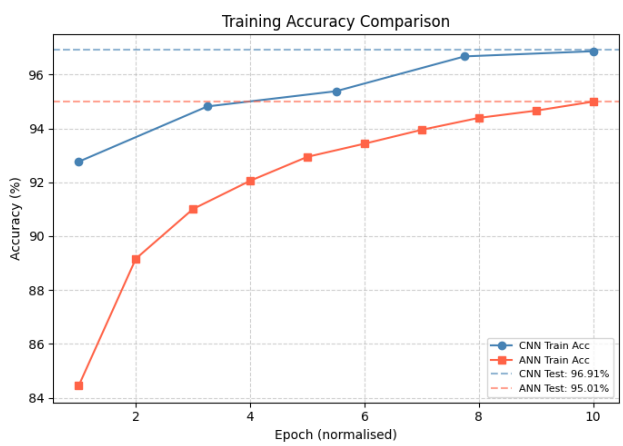

Figure 7: CNN vs ANN training and test accuracy comparison

5 Conclusion

- The CNN achieves higher accuracy than the ANN while using approximately 4.5× fewer parameters, demonstrating superior parameter efficiency.
- Weight sharing and local receptive fields significantly reduce the number of learnable parameters in CNNs.
- CNNs preserve spatial structure and learn hierarchical features, from low-level edges to high-level digit shapes.
- The ANN loses spatial locality due to flattening, requiring more parameters to learn similar representations.
- The CNN converges in fewer epochs but has higher computational cost per epoch.
- The ANN trains faster per epoch but requires more epochs and still achieves lower accuracy.
- For image-based tasks such as MNIST digit classification, convolutional architectures provide better generalization and efficiency than fully connected networks.